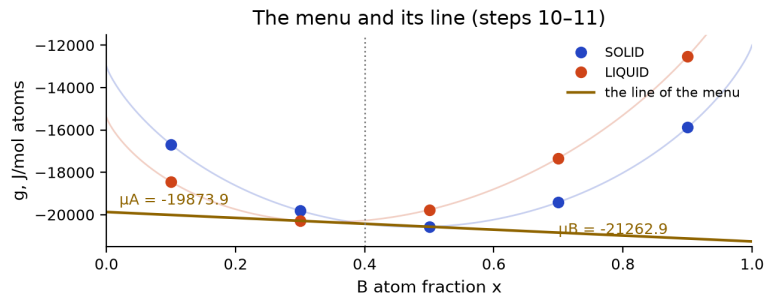
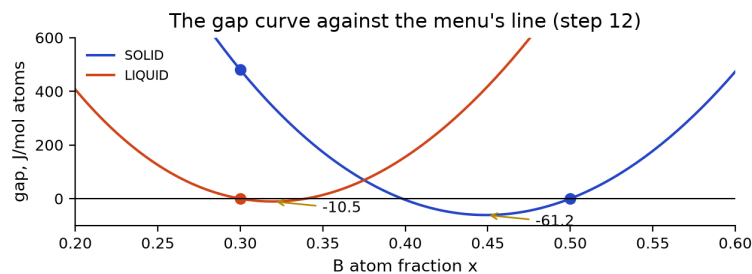


Advanced steps 07-18: the picture sheet

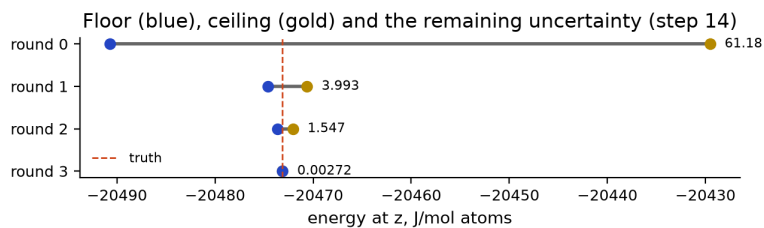
One page to keep beside the advanced steps 07-18. All energies are J/mol of atoms.



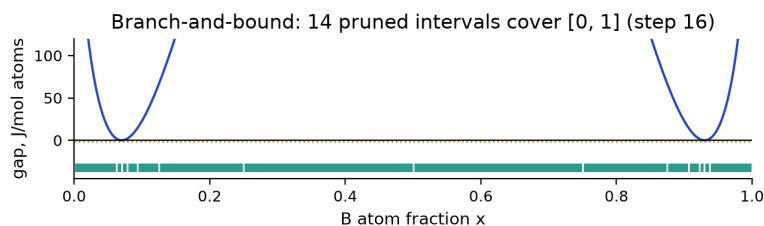
The menu and its line. At $z = 0.40$ the cheapest mixture of the menu uses LIQUID at 0.3 and SOLID at 0.5, at -20429.54 . The line through them lies under every dot: a floor for the menu. Its end heights are $\mu_A = -19873.9$ and $\mu_B = -21262.9$.



The gap curve. Energy minus the line. Used dots sit at zero; the curves dip below it between the dots, deepest at SOLID near 0.4489 (-61.2).



Ceiling and floor. The ceiling is the best mixture found; the floor is the line slid down by the deepest dip. The truth, -20473.12 , lies between them; their distance is the remaining uncertainty.



Branch-and-bound. On the regular solution at 800 K every interval of $[0, 1]$ has a floor of at least -1 against the final line: no valley deeper than the tolerance is left. Floor -5585.53 , ceiling -5585.11 .